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8 authors, including:

Sisi Tan
The University of Hong Kong

Jinghui Zou
Huazhong University of Science and Technology

Qiang Zhang
insititue of Metal Research , Chinese Academy of Sciences

yu yu
University of Reading

High-order all-optical differential equation solver based on microring resonators

Sisi Tan,[†] Lei Xiang,[†] Jinghui Zou, Qiang Zhang, Zhao Wu, Yu Yu, Jianji Dong, and Xinliang Zhang*

Wuhan National Laboratory for Optoelectronics & School of Optical and Electronic Information, Huazhong University of Science and Technology, Wuhan 430074, China

**Corresponding author: xlzhang@mail.hust.edu.cn*

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We propose and experimentally demonstrate a feasible integrated scheme to solve all-optical differential equations using microring resonators (MRRs) that is capable of solving first- and second-order linear ordinary differential equations with different constant coefficients. Employing two cascaded MRRs with different radii, an excellent agreement between the numerical simulation and the experimental results is obtained. Due to the inherent merits of silicon-based devices for all-optical computing, such as low power consumption, small size, and high speed, this finding may motivate the development of integrated optical signal processors and further extend optical computing technologies.

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For decades, optics has been proven to be the best means of conveying information. However, optical technology also has the opportunity to go beyond being just a convenient pipeline for ultrafast data transmission, and actually perform data processing and computing [1]. To accomplish the same goals as electronics, many computing technologies have been realized by optics, such as all-optical logics [2,3], optical differentiation [4–6] and integration [7–9], enabling the realization of more complex computing functions, such as solving differential equations.

Differential equations model and govern fundamental physical phenomena and engineering systems in virtually any field of science and engineering [10], such as temperature diffusion processes, physical problems of motion subject to acceleration inputs and frictional forces, and the response of different resistor–capacitor circuits, etc. Previous schemes have focused on two routes toward solving all-optical differential equations. The first one is based on an optical feedback loop [11,12], while the second one requires a properly designed temporal impulse response [13,14]. Nevertheless, the configurations of these schemes are bulky and complex, involving either a redundant loop [11,12] or an additional optical pump [12–14]. Meanwhile, schemes in [11–13] can only solve first-order ordinary differential equations (ODEs), but not high-order ones.

In this Letter, we experimentally demonstrate a simple all-optical linear ODE solver based on microring resonators (MRRs). The MRRs are designed with different radii and quality factors (Q), corresponding to solving first-order linear ODE with different constant coefficients. Employing single and cascaded MRRs, respectively, first- and second-order linear ODEs are solved by the cascaded MRR unit.

The common constant coefficient first-order linear ODE can be expressed as

$$\frac{dy(t)}{dt} + ky(t) = x(t), \tag{1}$$

where $x(t)$ represents the input signal, $y(t)$ is the output signal (equation solution), and k denotes a positive constant of an arbitrary value.

The traditional computing system [15] to solve Eq. (1) requires a feedback from the output. To implement such a physical system, one must introduce a loop-based structure, where a differentiator [4–6] or integrator [7–9] is indispensable.

The design of an all-optical ODE solver can also be approached by simply considering the response of the device in the frequency domain. The spectral transfer function $H(\omega)$ of an ideal filter to solve Eq. (1) can be described as

$$H(\omega) = \frac{1}{j\omega + k}, \tag{2}$$

where $j = \sqrt{-1}$, and ω is the optical frequency variable. From Eq. (2) one can see that an optical integrator is a special case of an ideal all-optical ODE solver when $k = 0$ [8]. Therefore, a promising approach for realizing an ideal all-optical ODE solver can be achieved by using an optical integrator slightly deviating from the lossless condition, under which $k \neq 0$.

According to the time-dependent relations and coupled mode theory, the transfer function at the drop port of an add–drop MRR can be given by [16]

$$T(\omega) = \sqrt{\frac{4}{\tau_{co}\tau_{tr}}} \Bigg/ \left[j(\omega - \omega_0) + \frac{1}{\tau} \right], \tag{3}$$

where ω_0 is the resonance frequency and τ is the photon lifetime. The reciprocal of photon lifetime is the decay rate, which is related to the power coupled to the transmitted wave, $1/\tau_{tr}$, the power lost due to the intrinsic effects, $1/\tau_{ie}$, and the power coupled to the output waveguide, $1/\tau_{co}$. Thus, $1/\tau = 1/\tau_{tr} + 1/\tau_{ie} + 1/\tau_{co}$.

From Eq. (3) it is obvious that the transfer function at the drop port of an add–drop MRR is in great agreement with the ideal filter [Eq. (2)] required to solve Eq. (1): $T(\omega) \propto 1/[j(\omega - \omega_0) + k]$. Therefore, a single add–drop MRR can be utilized to solve a constant coefficient first-order linear ODE, where the constant coefficient $k = 1/\tau$. Moreover, since the Q factor is decided by the photon lifetime [17], $Q = \omega_0 \tau / 2$, the constant coefficient can be expressed as $k = \omega_0 / 2Q$.

Figure 1 shows the feasibility diagrams of solving linear ODEs up to third order when cascading the corresponding number of MRRs. Figure 1(a) (Media 1) demonstrates the first-order linear ODE solver when utilizing the input and drop ports of a MRR. Then, the second-order linear ODE can be simply solved by cascading two MRRs when the output of the first MRR (MRR1) serves as the input of the second MRR (MRR2), as illustrated in Fig. 1(b), where the inputs and outputs of the two MRRs meet the condition that

$$\frac{dy_1(t)}{dt} + k_1 y_1(t) = x(t), \tag{4}$$

$$\frac{dy_2(t)}{dt} + k_2 y_2(t) = y_1(t). \tag{5}$$

Consequently, the output of MRR2 is the solution of a second-order linear ODE,

$$\frac{d^2y(t)}{dt^2} + a \cdot \frac{dy(t)}{dt} + by(t) = x(t), \tag{6}$$

where $a = k_1 + k_2$, $b = k_1 \cdot k_2$; k_1 and k_2 are the constant coefficients of the two first-order linear ODEs that are solved by MRR1 and MRR2, respectively. Similarly, a third-order linear ODE, expressed as

$$\frac{d^3y(t)}{dt^3} + a \cdot \frac{d^2y(t)}{dt^2} + b \cdot \frac{dy(t)}{dt} + cy(t) = x(t), \tag{7}$$

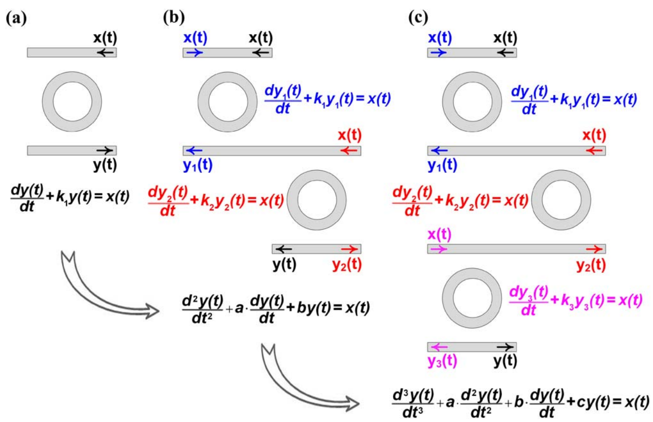

Fig. 1. (a) First-order linear ODE solver based on an add–drop MRR (Media 1); (b) first-order {blue [top left $x(t)$ and middle $y_1(t)$] and red [middle $x(t)$ and bottom $y_2(t)$ arrows]} and second-order {black [top $x(t)$ and bottom $y(t)$] arrows} linear ODE solver based on two cascaded MRRs; and (c) first-order {blue [left top $x(t)$, middle $y_1(t)$], red [middle right $x(t)$ and bottom right $y_2(t)$], and magenta [bottom left $x(t)$] arrows} and third-order {black [bottom $y(t)$] arrows} linear ODE solver based on three cascaded MRRs.

can also be solved when cascading three MRRs, as shown in Fig. 1(c), where $a = k_1 + k_2 + k_3$, $b = k_1 k_2 + k_1 k_3 + k_2 k_3$, $c = k_1 k_2 k_3$; and k_1, k_2 and k_3 are the constant coefficients of the three first-order linear ODEs that are solved by the three MRRs, respectively. Therefore, by cascading more MRRs, higher-order linear ODEs can also be solved accordingly.

We employed two cascaded MRRs fabricated on silicon-on-insulator (SOI) wafers with different radii ($R1 = 130$ um, $R2 = 100$ um) and Q factors ($Q1 = 19045$, $Q2 = 22038$), as illustrated in Figs. 2(a) and 2(b). The waveguide width and thickness of both straight and bending waveguides are 450 and 220 nm, respectively, while the gap between them is 200 nm [Fig. 2(c)]. The thickness of the top silicon and the buried oxide layers of SOI wafer are 220 nm and 3 μm, respectively. The device layout is then transferred to a ZEP520A photoresist by E-beam lithography (Vistec EBPG5000 + ES). Finally, the upper silicon layer is etched downward for 220 nm through the induced coupled plasma (ICP) etching (Oxford Instruments Plasmalab System 100). Theoretically, cascaded MRRs with the same radius can also be utilized to solve second-order differential equations. However, in practice, it is difficult to precisely match their spectra without a tuning technique. Therefore, two MRRs with different radii are utilized in our experiment, due to the Vernier effect. There is always one peak in the spectra of the two cascaded MRRs that perfectly matches and whose wavelength determines the experimental signal wavelength. The measured spectra of each MRR and the cascaded ones are illustrated in Fig. 2(d). The highest spectral peak of the cascaded MRRs is selected, which is at the wavelength of 1542.67 nm.

The experimental setup is shown in Fig. 3. The tunable laser diode (TLD) emits a continuous wave (CW) beam with a tuning resolution of 0.01 nm. The CW beam is then modulated by Mach–Zehnder modulators (MZM) driven by data signals generated from a bit pattern generator (BPG). Owing to the trailing of the output pulses, to better observe the output waveforms, the light is modulated by a BPG with 10 Gb/s self-coded data signals of a period

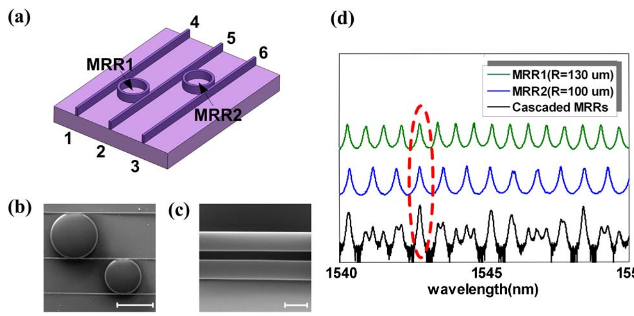

Fig. 2. (a) Schematic of the cascaded MRRs composed of two add–drop MRRs with different radii and Q factors (vertical grating couplers not shown); (b) SEM picture of the cascaded MRRs, and the scale bar represents 200 μm; (c) SEM picture of the coupling region between the ring and the straight waveguide, and the scale bar represents 500 nm; and (d) measured spectra of each MRR and the cascaded ones, employing ports 1 and 2 for MRR1 (green/top solid line), ports 5 and 6 for MRR2 (blue/middle solid line) and ports 3 and 4 for the cascaded MRRs (black/bottom solid line)

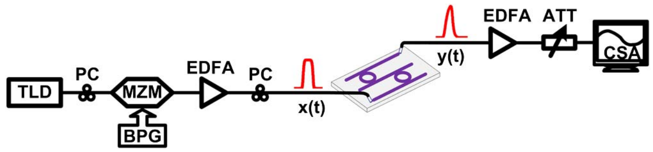

Fig. 3. Experimental setup for the high-order all-optical ODE solver with the on-chip MRRs.

of 1000, indicating a repetition rate of 2.5 GHz. The forward erbium-doped fiber amplifier (EDFA) is used to increase the optical power injected into the MRR, which is ~12 dBm (power ratio). A polarization controller (PC) is required since the silicon waveguide operates only in transverse electrical (TE) mode. The backward EDFA is used to compensate for the power loss by the chip. The integrated chip of MRRs is coupled to an optical fiber through vertical grating couplers. After power control by the backward EDFA and an attenuator (ATT), the output waveform at the drop port of MRR is analyzed through a communication signal analyzer (CSA). By utilizing different ports of the cascaded MRRs, first- and second- order linear ODEs can be solved accordingly.

Considering the most common Gaussian and super-Gaussian pulses as the input $x(t)$ in Eq. (1), respectively, the solutions achieved by the ideal filter are illustrated in Figs. 4(a) and 4(b). It can be readily seen that, for super-Gaussian input, the difference between the input and output is more observable. Therefore, we chose the super-Gaussian pulse in Fig. 4(c) as the experimental input, each with a full width at half-maximum (FWHM) of 100 ps at a repetition rate of 2.5 GHz.

As for Fig. 2(a), by employing ports 1 and 2 and ports 5 and 6, respectively, MRR1 and MRR2 are used as first-order all-optical ODE solvers, correspondingly with constant coefficients $k_1 = 0.032/\mathrm{ps}$ and $k_2 = 0.028/\mathrm{ps}$. The measured amplitude spectra, simulated phase spectra, and measured output waveforms are depicted in Figs. 5(a)–5(f), along with the theoretical results achieved by ideal filters. As the amplitude and phase spectra curves of the MRR and the ideal filter fit well, the output waveforms in Figs. 5(c) and 5(f) closely approximate the ideal solutions.

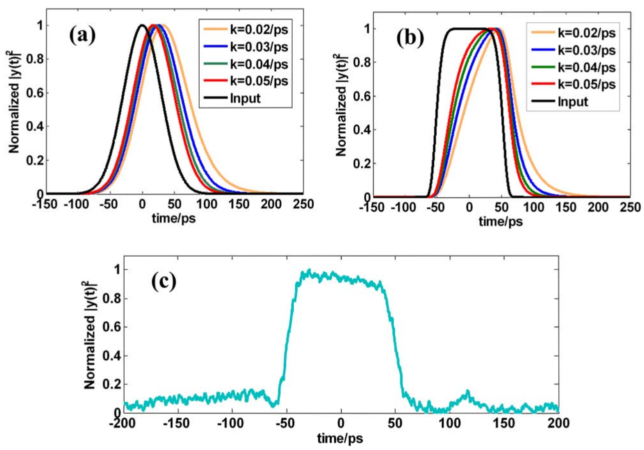

Fig. 4. Theoretical solutions of Eq. (1) with different constant coefficients for two kinds of input pulses: (a) Gaussian input; (b) super-Gaussian input; and (c) experimental input signal.

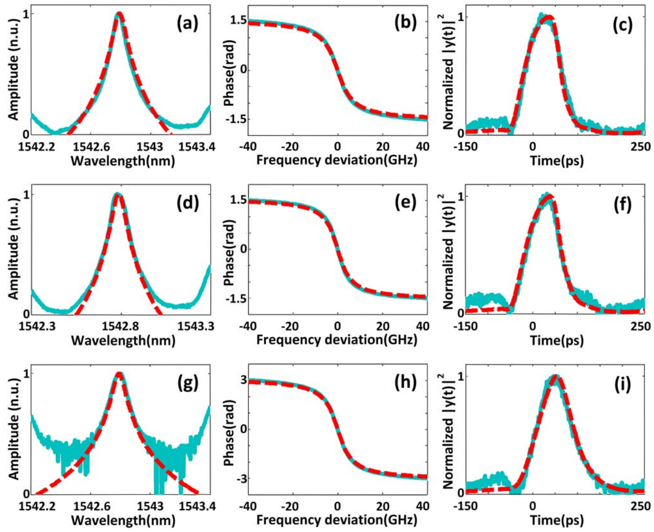

Fig. 5. Measured amplitude spectra, simulated phase spectra, measured output waveforms (turquoise/solid line), and corresponding simulated ideal results (red/dotted line) of the MRR-based all-optical ODE solver: (a) amplitude spectrum, (b) phase spectrum, and (c) temporal waveform for MRR1; (d) amplitude spectrum, (e) phase spectrum, and (f) temporal waveform for MRR2; (g) amplitude spectrum, (h) phase spectrum, and (j) temporal waveform for the cascaded MRRs.

Additionally, when using ports 3 and 4, the solution of Eq. (6) for $a = 0.06/\mathrm{ps}$ and $b = 0.000896/\mathrm{ps}^2$ can be obtained. The measured amplitude spectra, simulated phase spectra, and measured output waveforms are demonstrated in Figs. 5(g)–5(i) along with the ideal ones, which confirm the feasibility of using cascaded MRRs to solve high-order all-optical ODEs.

As illustrated in Fig. 2(d), because of the periodic frequency comb with a period fixed by the free spectral range, the response of a passive MRR deviates from the ideal all-optical ODE solver, particularly in the spectral region away from the resonance frequency which limits the bandwidth of the MRR-based all-optical ODE solver.

Additionally, as for other passive photonic devices, the overall energy efficiency of our all-optical ODE solver is fundamentally limited by the ratio of the resonator bandwidth to the signal spectral bandwidth [8]. Moreover, the practical throughput will be further reduced by the coupling and transmission losses. In practice, the propagation loss of the MRR is very low, whereas the total fiber to fiber loss is around 30 dB, mainly due to the loss of the vertical grating couplers and the coupling loss between the straight and bending waveguide. To increase the energy efficiency by introducing gain [18], for example, the energy efficiency could be improved. However, it comes at the expense of deteriorating the noise characteristics and complicating the configuration.

Moreover, the constant coefficient, k, is determined by the Q factor, which is related to the coupling coefficient between the straight and bending waveguides. Since tunability is essential for the system application of optical devices, in our case, aiming to realize a tunable all-optical equation solver with variable constant coefficients, one may introduce thermo-optic tuning or electro-optic tuning at the coupling region, such as employing a Mach–Zehnder based tunable coupler [19].

In summary, we demonstrate a simple scheme capable of solving first- and second-order linear ODEs based on MRRs. The scheme has the merits of compactness and expandability, due to the intrinsic advantages of MRRs, which provides a path for future fully integrated optical signal processors and computing systems.

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†These authors contributed equally to this work.

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